

Gauss-Seidel Iterative Method

Gauss-Seidel Iterative Method

$$\mathbf{Ax} = \mathbf{b}$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n = b_3$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = b_n$$

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Diagonally Dominant System:

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$$|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|$$

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$$|a_{11}| \geq |a_{12}| + |a_{13}| + \cdots + |a_{1n}|$$

$$\textcolor{red}{a}_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1$$

$$|a_{22}| \geq |a_{21}| + |a_{23}| + \cdots + |a_{2n}|$$

$$a_{21}x_1 + \textcolor{red}{a}_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2$$

$$|a_{33}| \geq |a_{31}| + |a_{32}| + \cdots + |a_{3n}|$$

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$$\vdots$$
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$$|a_{nn}| \geq |a_{n1}| + |a_{n2}| + \cdots + |a_{n,n-1}|$$

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$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \quad \rightarrow \quad x_1 = \frac{1}{a_{11}} \cdot (b_1 - a_{12}x_2 - a_{13}x_3 - \cdots - a_{1n}x_n)$$

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Recurrence Relations:

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Recurrence Relations:

$$x_1^{(k)} = \frac{1}{a_{11}} \cdot (b_1 - a_{12}x_2^{(k-1)} - a_{13}x_3^{(k-1)} - \cdots - a_{1n}x_n^{(k-1)})$$

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Gauss-Seidel Iterative Method

Example 03-06: Find root of the following system of equations using Gauss-Seidel iterative method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$\begin{array}{rcccccccl} & 10x_1 & - & x_2 & + & 2x_3 & & = & 6 \\ - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & & & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

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$$\begin{array}{rclclcl}
|10\rangle & \rangle & |-1\rangle + |2\rangle + |0\rangle = 3 & & 10x_1 & - & x_2 & + & 2x_3 & = & 6 \\
|11\rangle & \rangle & |-1\rangle + |-1\rangle + |3\rangle = 5 & - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 = 25 \\
|10\rangle & \rangle & |2\rangle + |-1\rangle + |-1\rangle = 4 & & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 = -11 \\
|8\rangle & \rangle & |0\rangle + |3\rangle + |-1\rangle = 4 & & & & 3x_2 & - & x_3 & + & 8x_4 = 15
\end{array}$$

Gauss-Seidel Iterative Method

Example 03-06: Find root of the following system of equations using Gauss-Seidel iterative method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$ and $\varepsilon_t = 10^{-6}$ for relative error.

$\mathbf{A}$ is diagonally dominant matrix.

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Recurrence Relations:

$$x_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})]$$

Gauss-Seidel Iterative Method

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Recurrence Relations:

$$\begin{aligned} x_1^{(k)} &= \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})] \\ x_2^{(k)} &= \frac{1}{11} \cdot [25 - (-x_1^{(k)}) - (-x_3^{(k-1)}) - (3x_4^{(k-1)})] \end{aligned}$$

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Gauss-Seidel Iterative Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

Gauss-Seidel Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = \frac{1}{10} \cdot \left[6 - (-x_2^{(0)}) - 2x_3^{(0)} \right]$$

Gauss-Seidel Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = \frac{1}{10} \cdot [6 - (-1) - 2 \cdot 1]$$

Gauss-Seidel Iterative Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$x_1^{(1)} = 0.5$$

Gauss-Seidel Iterative Method

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$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = \frac{1}{11} \cdot \left[25 - (-x_1^{(1)}) - (-x_3^{(0)}) - 3x_4^{(0)} \right]$$

Gauss-Seidel Iterative Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = \frac{1}{11} \cdot [25 - (-0.5) - (-1) - 3 \cdot 1]$$

Gauss-Seidel Iterative Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

Gauss-Seidel Iterative Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = \frac{1}{10} \cdot \left[-11 - 2x_1^{(1)} - (-x_2^{(1)}) - (-x_4^{(0)}) \right]$$

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$$x_3^{(1)} = \frac{1}{10} \cdot [-11 - 2 \cdot 0.5 - (-2.136364) - (-1)]$$

Gauss-Seidel Iterative Method

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$$x_1^{(1)} = 0.5$$

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$$x_3^{(1)} = -0.886364$$

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$$x_1^{(1)} = 0.5$$

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$$x_4^{(1)} = \frac{1}{8} \cdot [15 - 3x_2^{(1)} - (-x_3^{(1)})]$$

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$$x_4^{(1)} = 0.963068$$

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$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2}$$

Gauss-Seidel Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

Gauss-Seidel Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2}$$

Gauss-Seidel Iterative Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = \frac{1}{10} \cdot [6 - (-x_2^{(1)}) - 2x_3^{(1)}]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = \frac{1}{10} \cdot [6 - (-2.136364) - 2 \cdot (-0.886364)]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = \frac{1}{11} \cdot \left[25 - (-x_1^{(2)}) - (-x_3^{(1)}) - 3x_4^{(1)} \right]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = \frac{1}{11} \cdot [25 - (-0.990909) - (-(-0.886364)) - 3 \cdot 0.963068]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = \frac{1}{10} \cdot \left[-11 - 2x_1^{(2)} - (-x_2^{(2)}) - (-x_4^{(1)}) \right]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = \frac{1}{10} \cdot [-11 - 2 \cdot 0.990909 - (-2.019576) - (-0.963068)]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = \frac{1}{8} \cdot [15 - 3x_2^{(2)} - (-x_3^{(2)})]$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = \frac{1}{8} \cdot [15 - 3 \cdot 2.019576 - (-(-0.999917))] = 1.125$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(0.990909)^2 + (2.019576)^2 + (-0.999917)^2 + (0.992669)^2}$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(0.990909)^2 + (2.019576)^2 + (-0.999917)^2 + (0.992669)^2} = 2.654398$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(0.990909)^2 + (2.019576)^2 + (-0.999917)^2 + (0.992669)^2} = 2.654398$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.490909)^2 + (-0.116787)^2 + (-0.113554)^2 + (0.029601)^2}$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(0.990909)^2 + (2.019576)^2 + (-0.999917)^2 + (0.992669)^2} = 2.654398$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.490909)^2 + (-0.116787)^2 + (-0.113554)^2 + (0.029601)^2} = 0.518075$$

Gauss-Seidel Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(0.990909)^2 + (2.019576)^2 + (-0.999917)^2 + (0.992669)^2} = 2.654398$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.490909)^2 + (-0.116787)^2 + (-0.113554)^2 + (0.029601)^2} = 0.518075$$

$$\varepsilon^{(2)} = \frac{\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2}{\|\mathbf{x}^{(2)}\|_2} = \frac{0.518075}{2.654398} = 0.195176 > \varepsilon_t$$

Gauss-Seidel Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.136364$$

$$x_3^{(1)} = -0.886364$$

$$x_4^{(1)} = 0.963068$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.136364)^2 + (-0.886364)^2 + (0.963068)^2} = 2.554837$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.136364)^2 + (-1.886364)^2 + (-0.036932)^2} = 2.258551$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.258551}{2.554837} = 0.884030 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.136364, -0.886364, 0.963068)$$

$$x_1^{(2)} = 0.990909$$

$$x_2^{(2)} = 2.019576$$

$$x_3^{(2)} = -0.999917$$

$$x_4^{(2)} = 0.992669$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(0.990909)^2 + (2.019576)^2 + (-0.999917)^2 + (0.992669)^2} = 2.654398$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.490909)^2 + (-0.116787)^2 + (-0.113554)^2 + (0.029601)^2} = 0.518075$$

$$\varepsilon^{(2)} = \frac{\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2}{\|\mathbf{x}^{(2)}\|_2} = \frac{0.518075}{2.654398} = 0.195176 > \varepsilon_t \rightarrow \text{continue}$$